

University of Engineering and Technology, New Campus, Lahore
Department of Computer Science

Complex Computing Problem

Course Code & Title: CS-351 Machine Learning

1. Course Learning Outcome:

CLO 3: Design and develop end-to-end machine learning solutions for real-world problems by integrating data preprocessing, algorithm selection, and model tuning.

Level of CLO: CLO-3 is mapped to the 6th level, i.e., Create of Bloom's taxonomy.

2. CCP Attributes:

- **A2 Depth of analysis required:** Has no obvious solution and requires conceptual thinking and innovative analysis to formulate suitable abstract models.
- **A3 Depth of knowledge required:** A solution requires the use of in-depth computing or domain knowledge and an analytical approach that is based on well-founded principles.
- **A8 Interdependence:** Is a high-level problem possibly including many component parts or subproblems

3. Problem Statement:

Handwritten Urdu Alphabets Recognition System

Problem Description:

Design and implement a system that can recognize handwritten Urdu Alphabets from images captured in real-world conditions (e.g., mobile cameras, varying lighting, noisy backgrounds etc.). The images SHOULD be taken by each student himself / herself, and the Alphabets SHOULD have been written in his / her own handwriting. While testing, the system should accept images from any mobile phone camera.

4. Instructions:

- This is an individual project so each student will work independently.
- Your project will be graded according to the provided rubrics.
- Max Marks = 10
- **Deadline to submit: 25-04-2026**

5. Deliverables:

- Python code for data preprocessing, training, evaluation. (.txt file)
- Images collected in compressed / zip format
- 3-4 pages project report (PDF file)

6. Alignment with Attributes of Complex Computing Problems:

Attribute	Alignment / Justification
Depth of analysis required	Requires conceptual thinking and innovative analysis to find suitable ML models.
Depth of knowledge required	An analytical approach for comparing ML models and evaluating them.
Interdependence	Have subproblems such as data collection, data augmentation, model implementation etc.

Complex Computing Problem Assessment Rubrics

Criteria	Level 5 (Excellent)	Level 4 (Good)	Level 3 (Satisfactory)	Level 2 (Needs Improvement)	Level 1 (Poor/Incomplete)
1. Problem Understanding (A2,A3)	Demonstrates deep insight; problem scope well-defined	Good understanding with minimal ambiguity in scope and requirements.	Basic problem identified; partial clarity in scope or assumptions.	Problem vaguely defined; unclear or incorrect assumptions.	Problem not understood or requirements missing.
2. Data Preparation & Preprocessing (A3)	Data thoroughly cleaned and processed	Data is cleaned well; preprocessing is mostly appropriate.	Basic preprocessing done; some steps missing or not justified.	Limited preprocessing; inconsistent or incorrect handling.	No or incorrect preprocessing performed.
3. Model Selection & Implementation (A3, A8)	ML or DL models implemented correctly	All required models implemented and mostly appropriate.	2 models implemented but models poorly suited.	Only 1 model or incorrect implementation.	Missing or failed model implementation.
4. Evaluation & Metrics (A3, A8)	Excellent evaluation using multiple metrics	Good use of metrics; interpretation is mostly correct.	Metrics used but minimal analysis or misinterpretation.	Incomplete or shallow evaluation.	Metrics are missing or incorrect.
5. Comparative Analysis (A3, A8)	Deep, critical comparison of models; insightful trade-off discussion	Comparison includes relevant points; reasonably thorough.	Some comparisons are made, but lacks depth.	Limited comparison; mostly surface-level observations.	No or irrelevant comparison made.